Sparse, Layer-Aware Terrain Deformation for Open Worlds

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Abstract

A document exploring the proposal for a Doctoral Course at Tokyo Metropolitan University. An exploration of possible research directions and prior work as a foundation for studying terrain generation on existing game engines.

Keywords: Computer Graphics, Real-time, Game Engines, Terrain Deformation.

1 Summary and learnings

This is a proposal for the Doctoral Course at Tokyo Metropolitan University under the Mukai Laboratory, part of the Department of Industrial Art, Graduate School of Systems Design, Tokyo Metropolitan University and run by Professor Tomohiko Mukai.

The contents explore the idea of **Layered Terrain Deformation** applicable to Open Worlds. Much work has already been done on terrain rendering and deformation, but this proposal focuses on their practical applications. Usually linked to gameplay systems for games like Battlefield, Donkey Kong Bananza or Snowrunner, the objective is to present a generalised open-source solution that would be easy to plug and use in commercial engines like Unreal Engine or Unity. A **cross-engine abstraction layer** is being considered, but given the current market, it probably makes more sense to generalise the structure and shader code as much as possible while still creating engine-specific versions would be preferrable.

Systems that explore terrain deformation already exist, so one of the main differentiating points would be the existence of **Material Layers**. These would provide **extrapolations of current geological simulation rules** to deform soft terrain in a seemingly realistic manner. As a result of this deformation, procedural elements such as trails and paths could arise, providing a more believable world for users.

Lastly, this project aims to build from existing optimizations to create **a robust and fast architecture** that can be used on multiple platforms and systems with different specifications, making use of the power of **parallelised computing** on the GPU as much as possible.

2 Existing work

2.1 Overview

Terrain rendering and deformation has been an issue that has been studied for years. In this section there are multiple papers that explore rendering and deformation of terrains using GPUs. Data, **usually saved on textures**, can be optimised on different ways, like packing it into multiple textures to avoid loading on memory giant textures..[1] These height-maps can also be blended with directional noises to provide a better level of detail without needing high density

1. Dong Wang, Qing-Sheng Zhu, and Yi Xia, "Real-time multiresolution rendering for dynamic terrain," *J. Softw.* 9, no. 4 (April 2014).

textures for the deformation.[2] While the use of textures seems to be the norm, we can also find systems that use **hash-maps** to track which blocks or regions of the terrain (**height-map**) are affected by craters.[3]

Examples from 2007 illustrate how **terrain characteristics** like snow can be approximated via light calculations or ambient occlusion.[4] Very good insight on **Terrain Generation**[5] and **Terrain Descriptors**[6] helps understand how previous work on terrains has been done.

There has been some notable work done on non-real-time simulations using **velocity fields**[7] or, more precise and used for engineering purposes, **Material Point Methods**.[8] These simulate the behaviour of elements by describing it with **particles** that can be expressed and simulated on the GPU,[9] with examples of mixed material calculations for landslides.[10]

Lastly, there are some questions about whether an inter-engine Sparse Computing Middleware Software is worth developing. While there are definitely **uses of virtual caches on custom engines** like CryEngine,[11] Disney's Ptex,[12] and others based on lower level APIs,[13] there is no clear indication that such an abstraction layer is needed. Having a look at how **the majority of game sales are either Unreal, Unity or custom engines** and equally the majority of released games are made in Unity or Unreal Engine[14] the necessity becomes even less apparent. Still, there are some notable examples if we were to pursue this route of CUDA being connected to Unity,[15] C++ parallel frameworks[16] and abstractions for low level APIs.[17]

2. Giliam J P de Carpentier and Rafael Bidarra, "Interactive GPU-based procedural heightfield brushes," in *Proceedings of the 4th International Conference on Foundations of Digital Games* (Orlando Florida: ACM, April 2009).

3. Alfredo Cossetin Neto et al., "Real-time craters generation on dynamic terrains," in *Anais do XXIII Simpósio Brasileiro de Jogos e Entretenimento Digital (SBGames 2024)* (Brasil: Sociedade Brasileira de Computação, September 2024), 496–512.

4. David Foldes and Bedrich Benes, *Occlusion-Based Snow Accumulation Simulation*, 2007.

5. Guillaume Cordonnier et al., "Large scale terrain generation from tectonic uplift and fluvial erosion" [in en], *Comput. Graph. Forum* 35, no. 2 (May 2016): 165–175.

6. O Argudo et al., "Terrain descriptors for landscape synthesis, analysis and simulation" [in en], *Comput. Graph. Forum* 44, no. 2 (May 2025).

7. Balázs Jákó and Balázs Tóth, *Fast hydraulic and thermal erosion on GPU*, 2011.

8. Bingke Liu et al., "Study on large deformation of soil-rock mixed slope based on GPU accelerated material point method" [in en], *Sci. Rep.* 14, no. 1 (March 2024): 6983.

9. Hao Chen, Shiwei Zhao, and Jidong Zhao, "A sparse-memory-encoding GPU-MPM framework for large-scale simulations of granular flows" [in en], *Comput. Geotech.* 180, no. 107113 (April 2025): 107113.

10. Yi Zhang et al., "Material point method-based simulation of run-out characteristics for a soil–rock mixed landslide induced by surcharge" [in en], *Adv. Civ. Eng.* 2024 (February 2024): 1–12.

11. Martin Mittring and Crytek GmbH, "Advanced virtual texture topics," in *ACM SIGGRAPH 2008 Games* (Los Angeles California: ACM, August 2008).

12. Mark Lee, Nathan Zeichner, and Yining Karl Li, "A texture streaming pipeline for real-time GPU ray tracing," in *Proceedings of the Special Interest Group on Computer Graphics and Interactive Techniques Conference Talks* (Vancouver British Columbia Canada: ACM, August 2025), 1–3.

13. Javier Taibo, Antonio Seoane, and Luis Hernandez-Ibañez, "Dynamic virtual textures," *Journal of WSCG* 17 (January 2009); Cyril Crassin and Simon Green, "Octree-Based Sparse Voxelization Using the GPU Hardware Rasterizer," in *OpenGL Insights* (A K Peters/CRC Press, July 2012), 303–320; Igor Aherne et al., "Adoption of sparse 3D textures for voxel cone tracing in real time global illumination," in *Proceedings of the 15th International Joint Conference on Computer Vision, Imaging and Computer Graphics Theory and Applications* (Valletta, Malta: SCITEPRESS - Science / Technology Publications, 2020).

14. Dmitriy Byshonkov, *Video game insights: Game engines on steam in 2025*, February 2025, https://gamedevreports.substack.com/p/video-game-insights-game-engines.

15. David Algis et al., "InteropUnityCUDA: A tool for interoperability between Unity and CUDA" [in en], *Softw. Pract. Exp.* 55, no. 6 (June 2025): 1127–1141.

16. August Ernstsson, Lu Li, and Christoph Kessler, "SkePU 2: Flexible and type-safe skeleton programming for heterogeneous parallel systems" [in en], *Int. J. Parallel Program.* 46, no. 1 (February 2018): 62–80.

17. Robert Konrad and Stefan Göbel, "A cross-platform graphics API solution for modern and legacy development styles" [in en], in *Lecture Notes in Computer Science*, Lecture notes in computer science (Cham: Springer Nature Switzerland, 2023), 50–62.

2.2 Real Time Deformation

A System for Real-Time Deformable Terrain This research covers the development of a Real-Time Deformable Terrain System. Using a Graphical Processing Unit (GPU), the authors are able to write the deformation on a Height Map and a Partial Derivative Map to later be read by the Vertex and Fragment Shader. To optimise this work, the geometry for the clip-map is stored on the GPU via VBOs.[18]
https://pubs.cs.uct.ac.za/id/eprint/740/1/p77-crause.pdf

Real-time Multiresolution Rendering for Dynamic Terrain Work on how to properly display deformation **(Height Map - Depth Offset Map = Deformed Terrain Depth Texture)** tesselation[19]
https://www.jsoftware.us/vol9/jsw0904-14.pdf

Interactive GPU-based procedural heightfield brushes Exploration of a brush tool for terrain, using a **heightfield** and different noises for the brush strokes with directionality.[20]
https://www.decarpentier.nl/downloads/InteractiveGPUBasedProceduralHeightfieldBrushes_paper.pdf

Real-Time Craters Generation On Dynamic Terrains Optimized system, saved deformation craters to a **hashmap by compute shader** then read to check if a vertex is part of it.[21]
https://sol.sbc.org.br/index.php/sbgames/article/view/32332/32129

TRAIL: Simulating the impact of human locomotion on natural landscapes Closely related to this research. The work focuses on temporary physically-based footprints that damage vegetation and create tracks.[22]
https://link.springer.com/article/10.1007/s00371-024-03506-z

18. Justin Crause, Andrew Flower, and Patrick Marais, "A system for real-time deformable terrain," in *Proceedings of the South African Institute of Computer Scientists and Information Technologists Conference on Knowledge, Innovation and Leadership in a Diverse, Multidisciplinary Environment* (Cape Town South Africa: ACM, October 2011).
19. Wang, Zhu, and Xia, "Real-time multiresolution rendering for dynamic terrain."
20. Carpentier and Bidarra, "Interactive GPU-based procedural heightfield brushes."
21. Cossetin Neto et al., "Real-time craters generation on dynamic terrains."
22. Eduardo Alvarado et al., "TRAIL: Simulating the impact of human locomotion on natural landscapes" [in en], *Vis. Comput.* 40, no. 7 (July 2024): 5029–5041.

2.3 Erosion, Accumulation and Generation of Terrains

Occlusion-Based Snow Accumulation Simulation Old paper about snow accummulation by using average light direction and ambient occlusion.[23]
https://www.cs.purdue.edu/cgvlab/www/resources/papers/Foldes-2007-Occlusion-Based_Snow_Accumulation_Simulation.pdf

Terrain descriptors for landscape synthesis, analysis and simulation Article talking about different **Terrain Descriptors**. I am not sure how useful it would be for this, but it is state of the art and some things could be useful for certain characteristics like vegetation density from abiotic parameters.[24]
https://www.researchgate.net/publication/390799191_ Terrain_descriptors_for_landscape_synthesis_analysis_and_simulation

Large Scale Terrain Generation from Tectonic Uplift and Fluvial Erosion First time a terrain is generated by an Uplift Map with Fluvial Erosion. Not real time, but worth exploring. Convergence time with 10000 samples is 6.4s.[25]
https://inria.hal.science/hal-01262376v1

Fast Hydraulic and Thermal Erosion on GPU Old paper using **velocity fields** to calculate erosion on terrain. Good reference for **layers** as there is water and soil.[26]
https://diglib.eg.org/server/api/core/bitstreams/a5c270b1-da1c-498d-bb6d-5fa372ec54e0/content

A Sparse-Memory-Encoding GPU-MPM Framework for Large-Scale Simulations of Granular Flows Exploration on **Material Point Methods** for terrain simulation. Not usable for our real time cases.[27]
https://www.researchgate.net/publication/388231455_A_Sparse-Memory-Encoding_GPU-MPM_Framework_for_Large-Scale_Simulations_of_Granular_Flows

Material Point Method-Based Simulation of Run-Out Characteristics for a Soil–Rock Mixed Landslide Induced by Surcharge Similar article on **Material Point Methods** landslides. Not usable for our real time cases.[28]
https://www.researchgate.net/publication/378083721_Material_Point_Method-Based_Simulation_of_Run-Out_Characteristics_for_a_Soil-Rock_Mixed_Landslide_Induced_by_Surcharge

Study on large deformation of soil–rock mixed slope based on GPU accelerated material point method Similar to others, using **Material Point Method** to predict how large deformations could happen via GPU. Check for GPU Optimizations[29]
https://www.nature.com/articles/s41598-024-57362-x

23. Foldes and Benes, *Occlusion-Based Snow Accumulation Simulation*.
24. Argudo et al., "Terrain descriptors for landscape synthesis, analysis and simulation."
25. Cordonnier et al., "Large scale terrain generation from tectonic uplift and fluvial erosion."
26. Jákó and Tóth, *Fast hydraulic and thermal erosion on GPU*.
27. Chen, Zhao, and Zhao, "A sparse-memory-encoding GPU-MPM framework for large-scale simulations of granular flows."
28. Zhang et al., "Material point method-based simulation of run-out characteristics for a soil–rock mixed landslide induced by surcharge."
29. Liu et al., "Study on large deformation of soil-rock mixed slope based on GPU accelerated material point method."

2.4 Sparse Computing and GPU Optimizations

Advanced Virtual Texture Topics Exploration on how **virtual texture caches work** and their used on Crysis. Brings optimisations on memory access and enables new uses like decals or material blending.[30]
https://pdfs.semanticscholar.org/32f7/49c984e1ebd97e85f90d96b8ee5ed35c143a.pdf

Dynamic Virtual Textures Further optimisations on Virtual Textures for large landscapes. Worth investigating due to the improvements that it brought: efficient memory usage, several texture filtering... I'm specially interested on their tile loading setup and cache.[31]
https://naos-be.zcu.cz/server/api/core/bitstreams/43568dfc-6102-40c4-8fd9-77ed6e21820a/content

A Texture Streaming Pipeline for Real-Time GPU Ray Tracing Overview of Disney's new texture streaming pipeline for Ptex. Interesting general workings of the system.[32]
https://www.yiningkarlli.com/projects/gpuptex.html

Octree-Based Sparse Voxelization Using the GPU Hardware Rasterizer While the general paper talks about voxelisation, we can find some use for creating Sparse GPU programs.[33]
https://www.icare3d.org/research/OpenGLInsights-SparseVoxelization.pdf

Adoption of sparse 3D textures for voxel cone tracing in real time global illumination. Sparse 3D textures can serve as reference on their use for our deformation textures. It is also good reference on how the results are evaluated in terms of compute time at different resolutions.[34]
https://www.scitepress.org/Papers/2020/88745/88745.pdf

30. Mittring and Crytek GmbH, "Advanced virtual texture topics."
31. Taibo, Seoane, and Hernandez-Ibañez, "Dynamic virtual textures."
32. Lee, Zeichner, and Li, "A texture streaming pipeline for real-time GPU ray tracing."
33. Crassin and Green, "Octree-Based Sparse Voxelization Using the GPU Hardware Rasterizer."
34. Aherne et al., "Adoption of sparse 3D textures for voxel cone tracing in real time global illumination."

2.5 Engine interoperability

InteropUnityCUDA: A Tool for Interoperability Between Unity and CUDA Provides valuable insight on how to do compute shaders on the GPU, in this case with CUDA. While on our case we would aim for a more general OpenGL or DirectX solution, not locked to NVidia graphics cards, this is still very interesting.[35]
https://onlinelibrary.wiley.com/doi/epdf/10.1002/spe.3414

SkePU 2: Flexible and Type-Safe Skeleton Programming for Heterogeneous Parallel Systems Second version of the parallel C++ Framework SKePU, a framework centred towards heterogeneity and parallelism.[36]
https://link.springer.com/article/10.1007/s10766-017-0490-5

Diligent Graphics A Modern Cross-Platform Low-Level 3D Graphics Library and Rendering Framework focusing on working on many low level APIs and different platforms.[37]
https://link.springer.com/article/10.1007/s10766-017-0490-5

35. Algis et al., "InteropUnityCUDA: A tool for interoperability between Unity and CUDA."
36. Ernstsson, Li, and Kessler, "SkePU 2: Flexible and type-safe skeleton programming for heterogeneous parallel systems."
37. Ernstsson, Li, and Kessler.

3 Research Questions and Differentiating Aspects

The proposed Doctoral Research will be based on existing real-time, GPU Based, soft terrain deformation. As explored further, this research has been studied before and there are many solutions to this approach. However, many questions remain unanswered regarding **scalability, material realism, engine-integration and cross-platform compatibility.**

3.1 Research Questions

How can multi-layered terrain models (e.g., soil, gravel, bedrock) be represented and updated in real-time on the GPU.
Previous systems seem to be based on single terrain deformation. Soil or sand[38] and soft terrain[39] or simply generic terrain deformation.[40]

Can sparse GPU updates be integrated into virtualised rendering pipelines such as Unreal Engine's RVT?
(Under consideration) Runtime Virtual Textures caches shading data over large areas, usually used for decal-like effects suited to conform the terrain.[41] Can we provide a solution on how having an authoritative tiled texture atlas that then could be displayed and cached optimally with RVTs?

Can an engine-agnostic abstraction layer make real-time terrain deformation portable across different platforms (e.g., Unreal, Unity)?
(Under consideration) Unreal Engine and Unity are two engines used widely across the industry. While there are previous examples on adapting Graphics Systems like CUDA to commercial Game Engines[42] there hasn't been many uses of this. Mainly focusing on generalisation of low level APIs.[43] This would probably be overkill, and a common shader code adapted with specific plugins would suffice and would eliminate abstraction performance concerns.

How can localised deformation events be accumulated into a ground-truth texture atlas on the GPU, updating only dirty texels while retaining persistent history?
Optimisations would be necessary on such a large scale system. We would like to explore using a system that would only update dirtied texels, avoiding unnecessary redrawing and computing, as it could potentially scale on the number of deforming actors.

What kind of system would allow deformation events to accumulate forming paths or trails?
Repeated events would need to minimally physically or heuristically be modeled to allow this behaviour, and other similar results, to happen.

38. Alvarado et al., "TRAIL: Simulating the impact of human locomotion on natural landscapes."
39. Crause, Flower, and Marais, "A system for real-time deformable terrain."
40. Anthony S Aquilio et al., "Real-time GPU-based simulation of dynamic terrain," in *Advances in Visual Computing*, Lecture notes in computer science (Berlin, Heidelberg: Springer Berlin Heidelberg, 2006), 891–900.
41. Epic Games, *Runtime Virtual Textures*, Accessed: 2025-09-29 (Unreal Engine Documentation), https://dev.epicgames.com/documentation/en-us/unreal-engine/runtime-virtual-texturing-in-unreal-engine.
42. Algis et al., "InteropUnityCUDA: A tool for interoperability between Unity and CUDA."
43. Assiduous et al., *DiligentGraphics/DiligentEngine*, https://github.com/DiligentGraphics/DiligentEngine, October 2025, https://github.com/DiligentGraphics/DiligentEngine.

3.2 Differentiating Aspects

Layered Terrain Simulation.
Further exploration on deformation beyond single-heightfield deformation by supporting layers with different materials and properties when eroded.

GPU-First Architecture.
(Under consideration) By using a ground-truth atlas on the GPU, that could be saved onto the drive for persistent changes. Then, coupled with specific engine caching systems like RVT.[44]

Cross-Engine Middleware.
(Under consideration) Previously distracted work is developed on OpenGL independently. Proposes an abstraction layer not tied to a single engine, but also tied to existing game engines for easy portability and impact.

Content Generation.
Demonstrates how small changes can accumulate into larger patterns like paths or trails without explicit authoring.

44. Epic Games, *Runtime Virtual Textures*.

References

Aherne, Igor, Richard Davison, Gary Ushaw, and Graham Morgan. "Adoption of sparse 3D textures for voxel cone tracing in real time global illumination." In *Proceedings of the 15th International Joint Conference on Computer Vision, Imaging and Computer Graphics Theory and Applications*. Valletta, Malta: SCITEPRESS - Science / Technology Publications, 2020.

Algis, David, Berenger Bramas, Emmanuelle Darles, and Lilian Aveneau. "InteropUnityCUDA: A tool for interoperability between Unity and CUDA" [in en]. *Softw. Pract. Exp.* 55, no. 6 (June 2025): 1127–1141.

Alvarado, Eduardo, Oscar Argudo, Damien Rohmer, Marie-Paule Cani, and Nuria Pelechano. "TRAIL: Simulating the impact of human locomotion on natural landscapes" [in en]. *Vis. Comput.* 40, no. 7 (July 2024): 5029–5041.

Aquilio, Anthony S, Jeremy C Brooks, Ying Zhu, and G Scott Owen. "Real-time GPU-based simulation of dynamic terrain." In *Advances in Visual Computing*, 891–900. Lecture notes in computer science. Berlin, Heidelberg: Springer Berlin Heidelberg, 2006.

Argudo, O, E Guérin, H Schott, and E Galin. "Terrain descriptors for landscape synthesis, analysis and simulation" [in en]. *Comput. Graph. Forum* 44, no. 2 (May 2025).

Assiduous, Mikhail Gorobets, randompast, Andrey, Thomas TISSOT-DUPONT, Andrew H. Cox, Dzmitry Malyshau, et al. *DiligentGraphics/DiligentEngine*. https://github.com/DiligentGraphics/DiligentEngine, October 2025. https://github.com/DiligentGraphics/DiligentEngine.

Byshonkov, Dmitriy. *Video game insights: Game engines on steam in 2025*, February 2025. https://gamedevreports.substack.com/p/video-game-insights-game-engines.

Carpentier, Giliam J P de, and Rafael Bidarra. "Interactive GPU-based procedural heightfield brushes." In *Proceedings of the 4th International Conference on Foundations of Digital Games*. Orlando Florida: ACM, April 2009.

Chen, Hao, Shiwei Zhao, and Jidong Zhao. "A sparse-memory-encoding GPU-MPM framework for large-scale simulations of granular flows" [in en]. *Comput. Geotech.* 180, no. 107113 (April 2025): 107113.

Cordonnier, Guillaume, Jean Braun, Marie-Paule Cani, Bedrich Benes, Éric Galin, Adrien Peytavie, and Éric Guérin. "Large scale terrain generation from tectonic uplift and fluvial erosion" [in en]. *Comput. Graph. Forum* 35, no. 2 (May 2016): 165–175.

Cossetin Neto, Alfredo, Flávio Franzin, Cesar Tadeu Pozzer, Natan Berwaldt, Gabriel Di Domenico, and Gustavo de Freitas. "Real-time craters generation on dynamic terrains." In *Anais do XXIII Simpósio Brasileiro de Jogos e Entretenimento Digital (SBGames 2024)*, 496–512. Brasil: Sociedade Brasileira de Computação, September 2024.

Crassin, Cyril, and Simon Green. "Octree-Based Sparse Voxelization Using the GPU Hardware Rasterizer." In *OpenGL Insights*, 303–320. A K Peters/CRC Press, July 2012.

Crause, Justin, Andrew Flower, and Patrick Marais. "A system for real-time deformable terrain." In *Proceedings of the South African Institute of Computer Scientists and Information Technologists Conference on Knowledge, Innovation and Leadership in a Diverse, Multidisciplinary Environment*. Cape Town South Africa: ACM, October 2011.

Epic Games. *Runtime Virtual Textures*. Accessed: 2025-09-29. Unreal Engine Documentation. https://dev.epicgames.com/documentation/en-us/unreal-engine/runtime-virtual-texturing-in-unreal-engine.

Ernstsson, August, Lu Li, and Christoph Kessler. “SkePU 2: Flexible and type-safe skeleton programming for heterogeneous parallel systems” [in en]. *Int. J. Parallel Program.* 46, no. 1 (February 2018): 62–80.

Foldes, David, and Bedrich Benes. *Occlusion-Based Snow Accumulation Simulation*, 2007.

Jákó, Balázs, and Balázs Tóth. *Fast hydraulic and thermal erosion on GPU*, 2011.

Konrad, Robert, and Stefan Göbel. “A cross-platform graphics API solution for modern and legacy development styles” [in en]. In *Lecture Notes in Computer Science*, 50–62. Lecture notes in computer science. Cham: Springer Nature Switzerland, 2023.

Lee, Mark, Nathan Zeichner, and Yining Karl Li. “A texture streaming pipeline for real-time GPU ray tracing.” In *Proceedings of the Special Interest Group on Computer Graphics and Interactive Techniques Conference Talks*, 1–3. Vancouver British Columbia Canada: ACM, August 2025.

Liu, Bingke, Wen Wang, Zhigang Liu, Ningpeng Ouyang, Kejie Mao, and Fuchuan Zhou. “Study on large deformation of soil-rock mixed slope based on GPU accelerated material point method” [in en]. *Sci. Rep.* 14, no. 1 (March 2024): 6983.

Mittring, Martin, and Crytek GmbH. “Advanced virtual texture topics.” In *ACM SIGGRAPH 2008 Games*. Los Angeles California: ACM, August 2008.

Taibo, Javier, Antonio Seoane, and Luis Hernandez-Ibañez. “Dynamic virtual textures.” *Journal of WSCG* 17 (January 2009).

Wang, Dong, Qing-Sheng Zhu, and Yi Xia. “Real-time multiresolution rendering for dynamic terrain.” *J. Softw.* 9, no. 4 (April 2014).

Zhang, Yi, Fan Zhang, Qingbo Yi, Guanzhu Jiang, and Juncheng Xie. “Material point method-based simulation of run-out characteristics for a soil–rock mixed landslide induced by surcharge” [in en]. *Adv. Civ. Eng.* 2024 (February 2024): 1–12.